Object Oriented Programming (CSEG1044)

Unit 05 – Lecture 02: Set/SortedSet + Map/SortedMap

April 1, 2026

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Quick Links

Overview Key Topics Demo Summary

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Agenda

1 Overview

2 Key Topics

3 Demo / Example

4 Summary

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Lecture Snapshot

- **Unit focus:** Collections Framework and Wrapper Classes

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Lecture Snapshot

- **Unit focus:** Collections Framework and Wrapper Classes
- **Course thread:** Use Set/SortedSet + Map/SortedMap to manage in-memory project data more cleanly and predictably inside Campus Resource Manager.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Recall Before You Start

- **List basics**: Sets and maps extend the collection story by optimizing for uniqueness and key-based lookup. Main reference: `Unit 05 / Lecture 01 slides/notes`.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Recall Before You Start

- **List basics**: Sets and maps extend the collection story by optimizing for uniqueness and key-based lookup. Main reference: `Unit 05 / Lecture 01 slides/notes`.
- **Class design**: Set and map correctness depends on stable equality logic for stored objects. Main reference: `Unit 02 / Lecture 01 slides/notes`.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Learning Outcomes

- Choose Set vs List and `HashSet` vs `TreeSet`.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Learning Outcomes

- Choose Set vs List and `HashSet` vs `TreeSet`.
- Explain `equals`/`hashCode` and common mistakes.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Learning Outcomes

- Choose Set vs List and `HashSet` vs `TreeSet`.
- Explain `equals`/`hashCode` and common mistakes.
- Use `HashMap`/`TreeMap` for counting and sorted keys.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Key Terms to Know

- **Set**: collection with no duplicates.
- **SortedSet**: set that maintains sorted order (e.g., `TreeSet`).
- **Hashing**: converting an object to an int hash value (`hashCode`).
- **Map**: key-value association.
- **SortedMap**: map with sorted keys (e.g., `TreeMap`).
- **Contract**: if `a.equals(b)` then `a.hashCode() == b.hashCode()` must hold.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Study Flow for This Lecture

1. Refresh the prerequisite bridge before reading new ideas in isolation.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Study Flow for This Lecture

1. Refresh the prerequisite bridge before reading new ideas in isolation.
2. Study the main build in this order: *Set vs List; HashSet vs TreeSet* → *Equality and hashing basics (must know)* → *Map basics; HashMap vs TreeMap*.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Study Flow for This Lecture

1. Refresh the prerequisite bridge before reading new ideas in isolation.
2. Study the main build in this order: *Set vs List; HashSet vs TreeSet* → *Equality and hashing basics (must know)* → *Map basics; HashMap vs TreeMap*.
3. Trace the worked example and connect each line back to one concept frame.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Study Flow for This Lecture

1. Refresh the prerequisite bridge before reading new ideas in isolation.
2. Study the main build in this order: *Set vs List; HashSet vs TreeSet* → *Equality and hashing basics (must know)* → *Map basics; HashMap vs TreeMap*.
3. Trace the worked example and connect each line back to one concept frame.
4. Attempt the mini exercises before moving to the recap and exit check.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Unit 05: Collections and Wrapper Class

- Lecture focus: Set/SortedSet + Map/SortedMap

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Today: Roadmap

- Set vs list; ‘HashSet‘ vs ‘TreeSet‘

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Today: Roadmap

- Set vs list; ‘HashSet‘ vs ‘TreeSet‘
- Equality and hashing basics (‘equals‘/‘hashCode‘) and common mistakes

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Today: Roadmap

- Set vs list; ‘HashSet‘ vs ‘TreeSet‘
- Equality and hashing basics (‘equals‘/‘hashCode‘) and common mistakes
- Map basics; ‘HashMap‘ vs ‘TreeMap‘; common use-cases (counting/grouping)

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Set vs List

- List: ordered, duplicates allowed.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Set vs List

- List: ordered, duplicates allowed.
- Set: no duplicates; great for membership and deduplication.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

HashSet vs TreeSet

- `HashSet`: no guaranteed order; fast average operations.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

HashSet vs TreeSet

- `HashSet`: no guaranteed order; fast average operations.
- `TreeSet`: sorted order; needs `Comparable` or `Comparator`.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Why `equals` and `hashCode` matter

- Hash-based collections use `hashCode` to find a bucket.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Why `equals` and `hashCode` matter

- Hash-based collections use `hashCode` to find a bucket.
- `equals` confirms the match inside the bucket.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Why `equals` and `hashCode` matter

- Hash-based collections use `hashCode` to find a bucket.
- `equals` confirms the match inside the bucket.
- Contract: if `a.equals(b)` then `a.hashCode()==b.hashCode()`.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Map basics

- Map = key $\rightarrow$ value.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Map basics

- Map = key → value.
- Use cases: counting, grouping, lookups by id.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Map basics

- Map = key → value.
- Use cases: counting, grouping, lookups by id.
- `HashMap`: fast average; `TreeMap`: keys sorted.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Checkpoint and Reflection

Reflection prompt

Where would *Set/SortedSet + Map/SortedMap* matter most in Campus Resource Manager, and which design choice would you justify using this lecture's ideas?

Quick self-check

Which part needs one more example before you move on: *Set vs List; HashSet vs TreeSet*, *Equality and hashing basics (must know)*, the worked example, or the recap?

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Example: Set + Map counting

```
import java.util.*;
public class Demo {
    public static void main(String[] args) {
        List<String> names = Arrays.asList("Asha", "Ravi", "Asha
            ");
        Set<String> unique = new HashSet<>(names);
        System.out.println(unique);

        String[] words = {"a","b","a","c","b","a"};
        Map<String,Integer> freq = new HashMap<>();
        for (String w : words) freq.put(w, freq.getOrDefault(w
            ,0) + 1);
        System.out.println(freq);

        Map<String,Integer> sorted = new TreeMap<>(freq);
        System.out.println(sorted);
    }
}
```

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Mini Exercises (2–3 mins)

1. When do you prefer Set over List?

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Mini Exercises (2–3 mins)

1. When do you prefer Set over List?
2. Which keeps sorted order: `HashSet` or `TreeSet`?

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Mini Exercises (2–3 mins)

1. When do you prefer Set over List?
2. Which keeps sorted order: `HashSet` or `TreeSet`?
3. If you override `equals`, what else must you override?

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Watch-outs

- Forgetting the equals/hashCode contract and blaming the collection.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Watch-outs

- Forgetting the equals/hashCode contract and blaming the collection.
- Expecting HashSet or HashMap to preserve sorted order.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Watch-outs

- Forgetting the equals/hashCode contract and blaming the collection.
- Expecting HashSet or HashMap to preserve sorted order.
- Using a map or set before confirming the problem truly needs keys or uniqueness.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Key Takeaways

- Set enforces uniqueness; TreeSet provides sorted order.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Key Takeaways

- Set enforces uniqueness; TreeSet provides sorted order.
- Hashing relies on correct `equals`/`hashCode`.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Key Takeaways

- Set enforces uniqueness; TreeSet provides sorted order.
- Hashing relies on correct `equals`/`hashCode`.
- Map enables fast key-based lookup and counting/grouping.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Checkpoint Questions

1. Why do Hash-based collections need `hashCode` and `equals`?

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Checkpoint Questions

1. Why do Hash-based collections need `hashCode` and `equals`?
2. When do you choose `TreeMap` over `HashMap`?

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Checkpoint Questions

1. Why do Hash-based collections need `hashCode` and `equals`?
2. When do you choose `TreeMap` over `HashMap`?
3. Give one example of counting/grouping using a map.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Next Study Steps

- Revisit the recall references if any older idea still feels weak.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Next Study Steps

- Revisit the recall references if any older idea still feels weak.
- Rework the lecture example without looking at the notes first.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Next Study Steps

- Revisit the recall references if any older idea still feels weak.
- Rework the lecture example without looking at the notes first.
- Attempt the self-study practice and then answer the exit questions in full sentences.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Next Study Steps

- Revisit the recall references if any older idea still feels weak.
- Rework the lecture example without looking at the notes first.
- Attempt the self-study practice and then answer the exit questions in full sentences.
- Apply the idea once inside Campus Resource Manager to make the concept stick.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).

Next Lecture Preview

Next: Wrapped Collections + `Collections` utilities + Wrapper classes

- Utility methods and common pitfalls with autoboxing.

Sources: Schildt, *Java: The Complete Reference* (12e) (Collections Framework + wrapper classes).; Oracle Java docs: Collections Framework overview (accessed April 1, 2026).; Java SE API docs (`java.util` collections; wrapper classes in `java.lang`) (accessed April 1, 2026).